Olivier LEBLANC

Doctor in applied maths to computational imaging
UCLouvain

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Professional Experience

2025 **Post-doctoral Researcher**, UCLouvain, Belgium
Computational imaging methods for gravitational wave detection.

2025 **Sabbatical year**, Martinique & South America

2020–2024 **Doctoral Researcher**, UCLouvain, Belgium
Compressive and *neural-representation* strategies for inverse problems in *computational imaging* (reducing the computational resources towards real-time endoscopy and scalable radio-interferometry, enhancing diffraction tomography by plugging AI-based signal representations into state-of-the-art physical models). Supervisor: Laurent Jacques.

2020–2024 **Teaching Assistant (TA)**, for two courses at UCLouvain
- *Image Processing and Computer Vision* (∼50 students): designed, supervised, and graded various computer vision practical exams and exercise sessions in Python.
- *EE master's project* (∼50 students, 7 TAs + 4 professors): Designed and developed, for over 1.5 years, a prototype for a year-long autonomous audio-event classification project for environmental monitoring, co-led consultancy sessions, evaluations, and contests.

2018–2020 **Undergraduate TA**, coaching exercise sessions and grading homework
Signals and systems, electronics, electrical circuits, wave physics, and intro to quantum physics.

Software Contributor to Open-Source Python toolboxes on Github: BASPlib, pyunlocbox, pyproximal.

Scientific excellence Publications and oral presentations in the leading journals and conferences of the field, both as an autonomous researcher and as a team in international collaborations.

Reviewing Regular contributions to peer-review (IEEE TCI, Audio Engineering Society, ISCS23, COSERA24...).

Education

2020–2024 **PhD in Engineering Sciences and Technology**, UCLouvain, Belgium

2018–2020 **Masters degree in electrical engineering**, UCLouvain, *magna cum laude*
Areas: signal and image processing, information theory, cryptography, machine learning, telecommunications, optics, aerospace dynamics, nanoelectronics, electromechanical converters, robotics.

2015–2018 **Bachelors degree in electromechanical engineering**, UCLouvain, *cum laude*

Awards

Grant **Research Fellow**, *PhD funding for 4 years*, Fund for Scientific Research - FNRS, Belgium

First prize **Best contribution award**, *for my contributions to multicore fiber lensless imaging*, BASP Frontiers conference 2023, Villars-sur-Ollon, Switzerland

First prize **IEEE R8 Student Paper Contest 2021**, *for my experiments on walking droplets, TELSIKS 2021 conference*, Niš, Serbia

First prize **IEEE Best Master Thesis Award 2020**, *for my experiments on walking droplets*, UCLouvain

Professional Travels

2024 **Scientific stay**, *Two weeks of collaborative research on radio-interferometric imaging with the Biomedical and Astronomical Signal Processing (BASP) group led by Yves Wiaux*, Heriot-Watt University – Edinburgh, UK

2023 **Scientific stay**, *Two months of collaborative research on* implicit neural representations *with the Computational Imaging Group (CIG) led by U. Kamilov*, Washington University in St. Louis, USA

2022 **Scientific stay**, *One week of experimental measurements on an optical bench with a multicore fiber with the team of Hervé Rigneault*, Institut Fresnel – Aix-Marseille Université, France

2018 **Huawei**, *Two weeks to learn the ICT industry, research and suggest technologies to shape global sustainability efforts*, Beijing, Shenzen, and Hong Kong, China

Skills

Languages **French** (mother tongue), **English** (C2), Dutch (B1), Spanish (B1)

Programming **Python**, **Matlab**, **C**, Java

Parallel Programming **Pytorch**, **Matlab**, Cuda, Jax

Utilities LaTeX, HTML/CSS, git, Lt Spice, Labview, photo and video editing, 3D CAD, common office software

Outside of work

Personality Curious and loves learning, sociable, good-humoured, conscientious

Hobbies Science, sport, reading, building and repairing stuff, music

Peer-reviewed publications

Journal papers

[1] Olivier Leblanc, Mathias Hofer, Siddharth Sivankutty, Hervé Rigneault and Laurent Jacques, "*Interferometric lensless imaging: rank-one projections of image frequencies with speckle illuminations*", IEEE Transactions on Computational Imaging, vol. 10, pp. 208–222, February 2024

[2] Matteo Ravasi, Marcus Valtonen Örnhag, Nick Luiken, Olivier Leblanc, Eneko Uruñuela, "*PyProximal-scalable convex optimization in Python*", Journal of Open Source Software 9 (95), 6326, March 2024

[3] Murielle Kirkove, Yuchen Zhao, Olivier Leblanc, Laurent Jacques, Marc Georges, "*ADMM-inspired image reconstruction for terahertz off-axis digital holography*", JOSA A 41 (3), A1-A14, March 2024

[4] Olivier Leblanc, Yves Wiaux, Laurent Jacques, "*Compressive radio-interferometric sensing with random beamforming as rank-one signal covariance projections*", IEEE Transactions on Computational Imaging, vol. 11, pp. 1229–1242, July 2025

[5] Olivier Leblanc, Chung San Chu, Laurent Jacques, and Yves Wiaux, "*MROP: Modulated Rank-One Projections for compressive radio interferometric imaging*", Monthly Notices of the Royal Astronomical Society, vol. 543, n° 2, pp. 1727–1747, September 2025

Workshop abstracts

[6] Olivier Leblanc, Matthias Hofer, Siddharth Sivankutty, Hervé Rigneault, Laurent Jacques, "*An Interferometric view of Speckle Imaging*", Workshop on Low-Rank Models and Applications (LRMA), September 2022

Talks

* "*Towards Interferometric Lensless Endoscopy: Rank-one Projections of Images Frequencies with Speckle Illuminations*", Institut Fresnel's group, Marseille (France), February 2022
* "*Interferometric Lensless Imaging: Rank-one Projections of Image Frequencies with Speckle Illuminations*", Euler research building seminar, UCLouvain, February 2023
* "*My background and tutorial on diffraction imaging*", Computational Imaging Group, Washington University in St. Louis (USA), May 2023
* "*MultiCore Fiber Interferometric Imaging*", PoMM2024 : PhOtonique guidée Multimodale et Multi-canaux, Lille (France), May 2024